

OSCE-I

Emergency

Burn



- What do you see?
- Differential diagnosis ?
- Management ?

Cont'd

- What do you see?
- DDX
 - Burn (thermal, chemical, electrical, radiational)
 - SSSS (staphylococcal scalded skin syndrome)
 - Toxic shock syndrome
 - Steven Johnson syndrome
 - Kwashi dermatitis
 - cellulitis

Burn- mx

1. Emergency aid
2. Fluid resuscitation
3. Prevent hypothermia
4. Pain management
5. GI considerations* anti histamine
6. Antibiotics *not indicated in early post burn
7. TAT

Criteria for admission

- Full-thickness > 5% BSA
- Partial-thickness > 10% BSA
- Any full-thickness or partial-thickness burn involving **critical areas** (face, hands, feet, genitals, perineum, skin over major joint)
- **Children** with severe burns
- Circumferential burns of **thorax or extremities**
- Significant chemical injury, electrical burns, lightening injury, co-existing major trauma or significant pre-existing medical conditions
- Presence of inhalation injury

Burn- mx(2)

1. Emergency Aid

- ABC of life
- Avoid hypothermia
- Remove constricting clothes

2. Fluid resuscitation

- Vascular access: Two large bore peripheral lines (if possible) or central line.
- **4 ml R/L x % burn x body wt. In kg (parkland)**

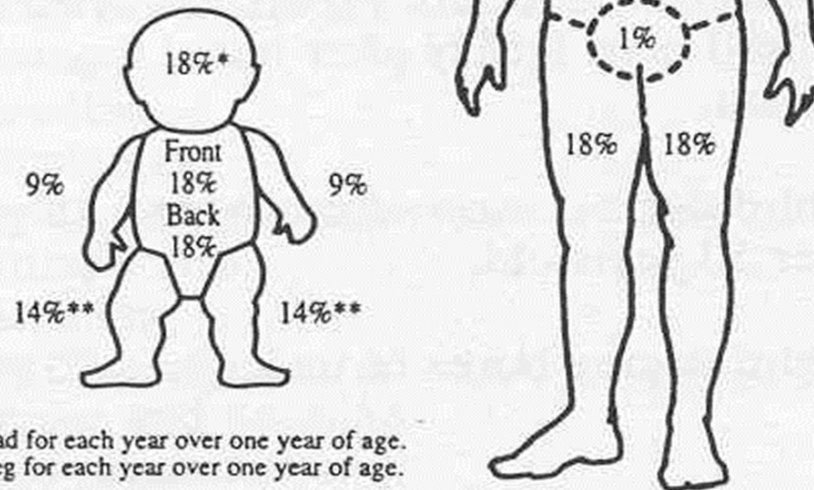
Cont'd

- ½ of calculated fluid is administered in the first **8 hours**
- Balance is given over the remaining **16 hours**.
- Maintain **urine output** at 0.5 cc/kg/hr.
- ARF may result from myoglobinuria
- Increased fluid volume, **mannitol bolus and NaHCO₃** into each liter of LR to alkalinize the urine may be indicated

Burn- Ass't

RULE OF 9's

(For calculating percentage of body burned)



Its inaccurate in childre

Burn- mx (2)...

3. Prevent hypothermia

- **Cover** patients with a dry sheet – keep head covered
- **Pre-warm** trauma room
- Administer warmed IV solutions
- **Avoid** application of saline-soaked dressings
- **Avoid** prolonged irrigation
- **Remove** wet / bloody clothing and sheets
- **Avoid** application of antimicrobial creams
- **Continual monitoring** of core temperature

Cont'd

4. Pain management

5. GI considerations

- **NGT** insertion to reduce potential for aspiration and paralytic ileus.
- Early administration of **H2 histamine receptor** recommended

6. Antibiotics

- Prophylactic antibiotics are not indicated in the early postburn period.

7. Other

- **TAT**
- **Debride and treat**

other

- SSSS
 - Cause: s.aures
 - Affect infants and children
 - Flexural accentuation and mucus membrane is involved
 - Mx: clindamycin, penicillin
- TSS
 - Cause: s.aures, GAS
 - Have associated multi organ failure unlike SSSS
 - Desquamation of hands, soles.. 1wk after the onset of acute illness
 - Mx: clindamycin

Cont'd

- Steven Johnson syndrome
 - Cause
 - Infections
 - Drug induced
 - Malignancy related idiopathic
- Mx: symptomatic rx
 - ABC of life
 - Withdrawal of the agent

Childhood Poisoning

Poisoning

- Organophosphates and Carbamates
 - Most commonly used insecticides
- c/f
 - ❖ *smells of Garlic odor and her clothes are soaked with vomit*
 - Nicotinic effects=**MATCH**
 - Muscle weakness
 - Adrenal medulla action
 - Tachycardia
 - Cramping of the skin
 - Hypertension
 - Hypoventilation
- Muscarinic = **DUMBBELS**
 - Diarrhea/defecation
 - Urination
 - Miosis
 - Bronchospasm
 - Bradycardia
 - Emesis
 - Lacrimation
 - Salivation

OP poisoning- mx

- Dx:
 - RBC cholinesterase and pseudocholinesterase concentrations can be measured
 - Response to atropine & pralidoxime??
- Mx:
 1. Supportive
 2. GI decontamination
 3. Dermal deconta.
 4. Anti dote admi.

Cont'd

1. Supportive

- ABC of life
- Fluid & electrolyte replacement
- Intubation/ ventilation

2. GI decontamination

- Activated charcoal
- lavage

3. Dermal deconta.

- Removal of clothing
- Wash with soap & water

4. Antidote

- Atropine
- Pralidoxime *not recommended in carbamate

Hydrocarbon poisoning

- Due to ingestion or inhalation of
 - Kerosine
 - Gasoline
 - Lamp oil
 - Paint thinners
- c/f
 - Aspiration pneumonia
 - Common manifestation
 - Cough; may progress to resp. failure
 - Fever and leukocytosis
 - Mild CNS depression

HC poisoning- mx

- Dx:
 - Hx & physical
 - Abnormal CXR
- Mx:
 1. ABC of life
 2. Oxygen
 - Mechanical ventilation and ECMO
 3. Fluid and Electrolyte mx
 4. β -blockers (usually esmolol)
 - ❖ Emesis & lavage are **contraindicated**
 - ❖ Activated charcoal- **not useful**

Ethanol poisoning

- A history of ingestion from the caregiver or patient and characteristic physical findings, such as sickly sweet breath odor, altered mental status, and nystagmus with lateral gaze typically are sufficient to establish the diagnosis of ethanol intoxication. Detection of ethanol on the breath or in the serum provides additional confirmation.

Cont'd

- c/f:
 - exhilaration
 - loss of inhibition, which progresses to lack of coordination, ataxia, slurred speech, gait disturbances, drowsiness, and, ultimately, stupor and coma.
 - Mild to moderate metabolic acidosis
 - Hypothermia, bradycardia, hypotension, resp. distress
 - Irritation of GI mucosa
 - Inhibition of antidiuretic hormone action in the kidney

Ethanol poisoning- mx

- Ix:
 - RBS
 - Etahnol level
 - other ivestigations for child abuse, sexual abuse or trauma
- Mx:
 1. **Supportive**
 2. naloxone 2 mg IV push is indicated to rule out opiate toxicity * if no response to supportive
 3. Gastric decontamination
 - usually not needed unless child is comatose
 - Done in 1hr of ingestion
 4. hemodialysis
 - increases ethanol clearance by 3 to 4 times indicated with ethanol levels >500 mg/dL or hepatic dysfunction

Caustic poisoning

- Alkali burns
 - Liquifactive necrosis
 - Has deeper penetration
 - Due to ingestion of detergents, battery cell
 - **cause** perioral, oropharyngeal, and upper esophageal injury
- Acid burns
 - coagulative necrosis
 - Eschar formn. Avoids deeper penetration
 - **cause** severe gastric injury, including perforation

Cont'd

- c/f:
 - highly alkaline laundry detergents, can cause **life-threatening airway edema**
 - excessive crying
 - Drooling
 - refusal to eat or drink
- Mx:
 1. ABC of life
 2. Diluents
 3. Steroids
 - decrease the incidence of esophageal strictures in patients with severe burns

- **Acetaminophen poisoning**

- c/f :

Stage	Time after ingestion	Characteristics
I.	½ - 24 hr	Anorexia, nausea, vomiting, malaise, pallor, diaphoresis.
II.	24 - 48 hr	Resolution of above; RUQ abdominal pain and tenderness; elevated bilirubin, PT, hepatic enzymes; oliguria.

- IX:
 - plasma level after 4hr
 - LFT and enzymes
 - Mx:
 - activated charcoal
 - Antidote: N acetyl cystin

III.	72 - 96 hr	Peak liver function abnormalities; anorexia, nausea, vomiting, malaise may reappear.
IV.	4 day – 2 wk	Resolution of hepatic dysfunction or complete liver failure.

Cont'd

- Salicylates

- c/f

- Nausea, vomiting
 - DHN, met acidosis
 - agitation, confusion
 - Coma
 - Pulm. Edema
 - hypoglycemia

- Ix: sr level

- Urine PH
 - RBS
 - Electrolytes
 - LFT

- mx:

- rehydration & correct electrolyte abn.
 - Activated charcoal
 - Alkalization of urine via IV bicarbonates
 - Hemodialysis if sever

CO poisoning

- **c/f:** depend on the measured carboxy Hb level
- 10-20% headache, dyspnea on exertion
- 20-30% severe headache, nausea
- 30-40% severe head ache, dyspnea, impaired judgement
- 50-60% confusion, syncope, seizure
- >70% coma, cardioresp. distress, death
- **Dx**
 - Carboxy Hb level
 - CXR 30% abnormal
 - ECG
- **Mx**
 - Supportive
 - 100% oxygen by mask
 - Hyperbaric oxygen therapy

Management of drowning

- ABC of life
- Remove wet clothing
- Administer warmed and humidified 100% oxygen
- Monitor core temperature, heart and respiratory rates, and blood pressure continuously
- Consider central venous pressure determination for severe hypothermia

ETAT

Triage Categories

- All sick children are rapidly examined on arrival and sorted into 3 categories:
- **EMERGENCY** signs – who require immediate emergency treatment
- **PRIORITY** signs – who should be given priority in the queue, rapidly assessed and treated without delay
- **NON-URGENT** cases – who can wait their turn in the queue. The majority of children seen will be non-urgent cases.

- **Emergency signs** are sorted in order of priority as:
 - A = Airway
 - B = Breathing
 - C = Circulation, Coma, Convulsions
 - D = Dehydration
 - O= Others (bleeding child, trauma with open fracture, Acute poisoning and Exposing the child after poly trauma also is an emergency assessment)

- **Priority signs – 3 TPR MOB**
 - Tiny baby
 - Temperature
 - Trauma
 - Pallor
 - Poisoning
 - Pain
 - Restless
 - Respiratory distress
 - Referral
 - Malnutrition/ marasmus
 - Oedema
 - Burns

Cont'd

- **Non-urgent**

- Once assessment is complete if no emergency or priority signs are found the child is classed 'non-urgent' and should wait their turn in the queue

Airway and Breathing

- Is the child breathing?
- Is the breathing obstructed?
 - Tongue flopping back
 - Foreign body
 - Croup
 - Anaphylaxis
- history of choking
- chin lift with head tilted to appropriate position
- Airway positions
 - **Infant** <12 months – NEUTRAL (nose up)
 - **Child** >12 months – SNIFFING (chin up)



Figure 5
Sniffing position to open up airway in an older child ("chin up")

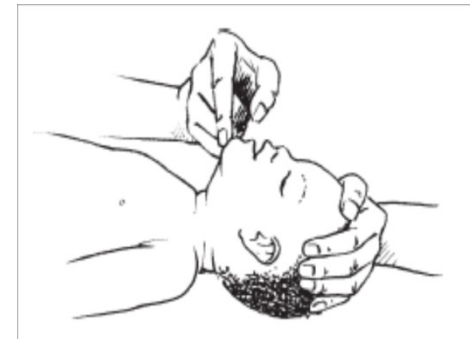
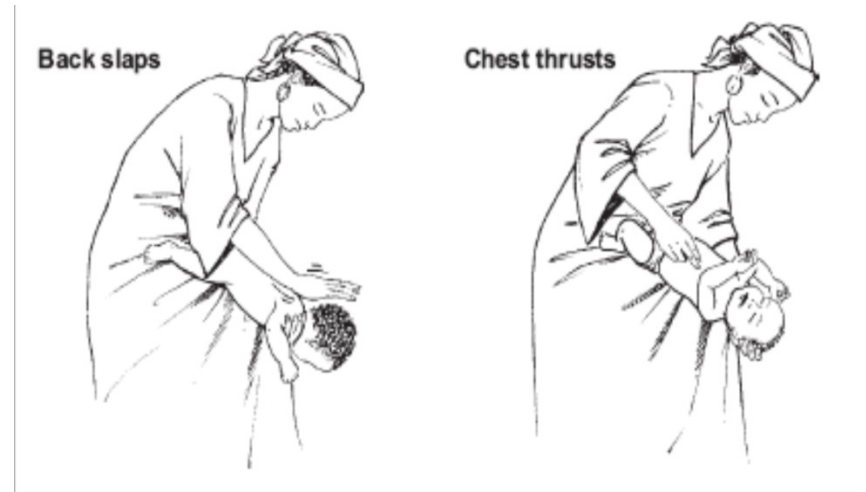


Figure 4
Neutral position to open airway in an infant ("nose up")

Management of the choking infant

- Lay infant on your arm in head down position
- Give 5 blows to the infants back with heel of hand
- If obstruction persists turn over and give 5 chest thrusts with 2 fingers, 1 finger breadth below nipple level in midline
- If obstruction persists check infants mouth
- If necessary repeat sequence with back slaps again



Management of Choking Child

- Give 5 blows to the child's back with heel of hand with child sitting, kneeling or lying
- If obstruction persists, go behind the child and pass your arms around the child's body: form a fist with one hand immediately below the child's sternum; place the other hand over the fist and pull upwards into the abdomen; repeat **this Heimlich manoeuvre** 5 times
- If the obstruction persists, check the child's mouth
- If necessary repeat sequence with back slaps again

Slapping the back to clear airway obstruction in a choking child



Heimlich manoeuvre in a choking older child



Figure 3
Management of child

Management of Airway in suspected head/neck trauma

- Avoid moving the head or neck
- Use JAW THRUST to open airway
- Stabilise the neck
- Keep lying on back with neck stabilised
- Tape forehead to sides of a firm board to secure position
- Support head on either side to prevent movement
- Place strap over chin
- If vomiting turn on side keeping head in line with body (LOG ROLL)



Figure 6
Jaw thrust without head tilt



Figure 7
Stabilising the neck when trauma is suspected

Insertion of an oropharyngeal (Guedel) Airway

- Can be used in unconscious patient to improve airway opening
- Not tolerated in patients who are awake (gag reflex)
- Infants– insert convex side (right way) up
- Children- insert concave side up (upside down) then turn
- Give Oxygen



Figure 14
Inserting an oropharyngeal airway in an infant: convex side up

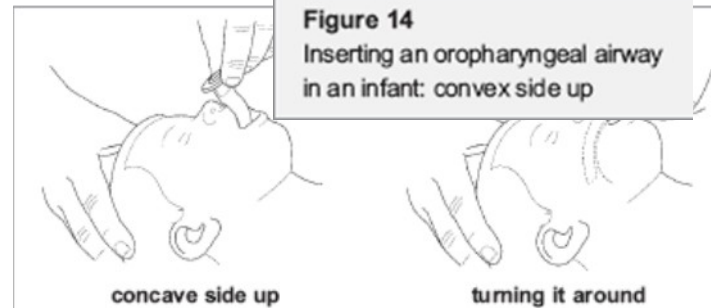


Figure 15
Inserting an oropharyngeal airway in an older child

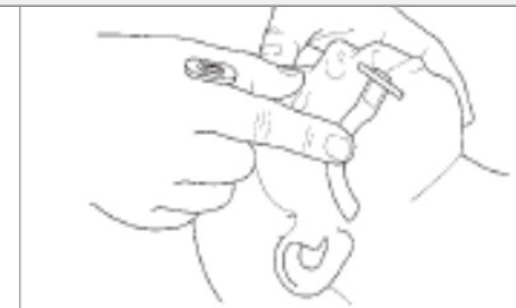


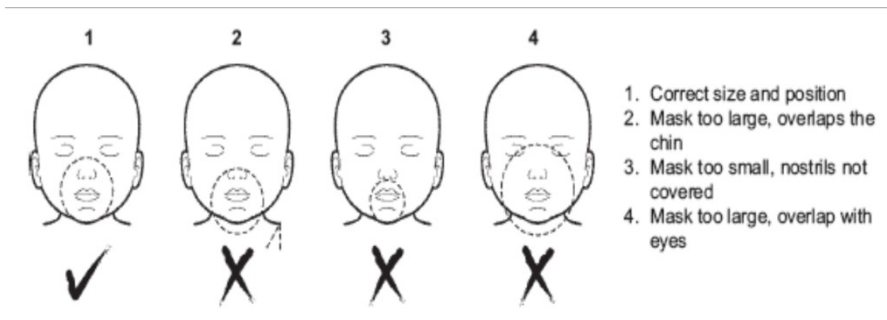
Figure 13
Selecting the right size of an oropharyngeal airway

Assessment of Breathing

- Is the child breathing?
- LOOK
 - Active, talking, crying = breathing
 - Chest moving?
- LISTEN
 - Listen for breath sounds
 - Are they normal or is breathing noisy/ obstructed?
- FEEL
 - Can you feel breath at the nose/ mouth of the child
- Signs of severe respiratory distress
 - Very fast breathing
 - Severe chest wall indrawing
 - Using accessory muscles/ head bobbing
 - Stridor
 - grunting

Management of Breathing Problems

- Ventilate with Bag and Mask
- Consider Oro-pharyngeal (Guedel) airway



Bag and mask

- child who is not breathing or is gasping
- Check bag is in good working
- Choose correct mask size
- Ensure good seal around mouth/nose and open airway
- Squeeze bag slowly and evenly
- Watch chest rise and allow to fall before giving next breath

Cont'd

- Sources of oxygen
 - Oxygen concentrator
 - Oxygen cylinders
- Delivery of oxygen
 - Nasal prongs
 - Nasal catheter
 - Face Mask
- Flow rates
 - Infant – 0.5-1litres/min
 - Child – 1-2 liters/min
 - Use flow rate of 5-10 lits/min for Face Mask
- Nasal prongs
 - Keep clear of mucus
 - Secure with tape
 - Use correct size or cut to fit
- Nasal catheter
 - 6-8 FG tubing
 - Insert to a distance= to nostril to inner eyebrow
 - Should not be visible below uvula
 - Avoid in pertussis or croup

Circulation

- Does the child have warm hands?
 - If not, is the capillary refill time longer than 3 seconds?
 - And is the pulse weak and fast?
 - Check the radial pulse
 - In infants brachial or femoral pulse may be easier
 - In older children carotid pulse may be felt
- ❖ If the pulse is weak and fast the child needs treatment for shock

Management of Shock

- A child with **cold hands, Cap. refill >3 secs, fast, weak pulse** is in shock
- Stop any bleeding
 - apply firm direct pressure, do not use tourniquet
- Give oxygen
- Make sure the child is warm
- Establish IV or intraosseous access
- Take blood samples for emergency lab tests
- Give fluids
 - IV/IO access
 - Give 20ml/kg Ringer's lactate or normal saline as fast as possible
 - Reassess circulation
 - ?No improvement
 - Repeat 20ml/s kg Ringers/Saline
 - Reassess
 - ?No improvement
 - Repeat 20ml/s kg Ringers/Saline
 - Reassess
 - ?No improvement
 - Give 20ml/kg blood over 30 mins OR
 - In severe diarrhoea repeat Ringers
 - Reassess
 - ?No improvement
 - Treat underlying condition

Management of shock in children WITH severe malnutrition

- Weigh the child
- IV fluid(Ringer/Ns with 5%glucose) 15ml/kg over 1hr
- Check pulse and breathing rate every 5-10mins
- **Stop** IV infusion if either of these; increase
 - pulse by 15,
 - RR by 5
- If there is improvement (pulse and breathing rate fall)
 - Repeat 15ml/kg over 1 hour
 - Switch to oral or NG ReSoMal 10ml/kg/h
- If there is NO improvement
 - Call senior health worker
 - Give maintenance IV fluid 4ml/kg /h while waiting for blood
 - Transfuse fresh whole blood at 10ml/kg/h slowly over 3 hours (use packed red cells if cardiac failure)

Intravenous Access

- Use if IV access impossible
- Avoid if:
 - Infection at puncture site
 - Fracture of the bone
- Site:
 - Proximal tibia 1-2cm below and medial to the tibial tuberosity
 - Alternative – distal femur, 2cm above lateral condyle
- Can use special needle or large needle (21g) + syringe
- Should be removed after 8 hours or as soon as IV access obtained

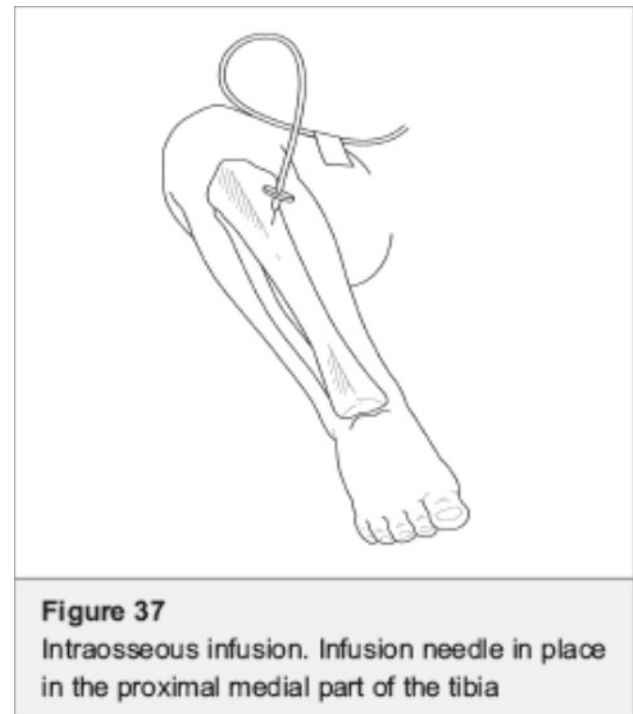


Figure 37

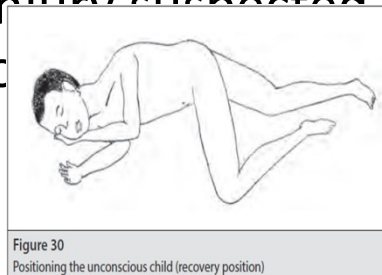
Intraosseous infusion. Infusion needle in place in the proximal medial part of the tibia

Coma and Convulsion

- A –Is the child ALERT? If not,
- V –Does the child respond to Voice? If not,
- P –Does the child respond to Pain?
- U – Child who is Unresponsive to Voice and Pain is Unconscious
- Any child with a coma score of 'P' or 'U' needs emergency treatment for coma
- Level of consciousness can also be assessed by GCS (for follow up of the child progress)

Management of Coma and Convulsion

- Airway
 - Keep open
 - Do not insert anything into mouth of convulsing child
- Position
 - Recovery position – turned to side, hand supporting head, bent lower leg maintains position
 - If spinal injury suspected stabilise cervical spine to prevent vomiting
- Check blood sugar
 - Dextrostix or glucometer
- If glucose $<45\text{mg/dl}$ (or 54mg/dl in a malnourished child) give dextrose
 - 5ml/kg 10% dextrose
- Recheck glucose after 30 mins – if still low repeat
- Feed as soon as conscious



Management of Convulsion

- If child convulsing Now give:
 - PR Diazepam 0.1ml/kg (10mg/2ml solution)
 - Or IV Diazepam 0.05ml/kg
- **NB Do not give Diazepam IM!**
- If still convulsing after 10 mins repeat diazepam (PR or IV).
After the 3rd dose give:
- Phenytoin 20mg/kg IO/IV over 20min (monitor for Arrhythmia)
- Phenobarbitone 15-20mg/kg IV or IM (for neonates)
- Reduce fever if present with tepid sponging

Management of status epileptics

causes

- **Definition**

- **continuous seizure** activity or **recurrent seizure** activity without regaining of consciousness lasting for more **than 5 min**

- CNS infections
- Acute hypoxic ischemic insult
- Metabolic disease (eg, hypoglycemia, inborn error of metabolism)
- Electrolyte imbalance
- Traumatic brain injury
- Drugs, intoxication, poisoning
- Cerebrovascular event

Cont'd

- Principles of management
 1. Treat life threatening conditions
 2. Treat the acute seizure
 3. Treat the underlying cause
 4. Prevent recurrence
 5. Prevent and treat systemic complications
- ABC...of life
- Diazepam Iv or rectal
- Phenytoin /fosphenytoin iv
 - Loading dose 20mg/kg
 - Reload with 10mg/kg
- Phenobarbital Iv/po loading dose 20 mg/kg
 - Reload 10 mg /kg
- Midazolam dripe
- Propofone
Thiopental ,pentobarbital

Febrile seizure

- Occur B/n age of 6 and 60 mo
- With a T^o of $\geq 38^{\circ}\text{C}$ (100.4°F) ,
- Are not the result of CNS infection or any metabolic imbalance
- Occur in the absence of a history of prior afebrile seizures.
- Febrile status epilepticus is a febrile seizure lasting longer than 30 min.

Dehydration

● NO severe malnutrition

- Give Ringer's Lactate
- Infants:
 - 30mls/kg over 1 hour
 - 70mls/kg over next 5 hours
- Children >1 year:
 - 30mls/kg over 30min then
 - 70mls/kg over 2½ hours
- Assess child every 1-2 h
- If not improving:
 - Give fluid more rapidly
 - Inform senior staff
- As soon as child can drink
 - Give ORS 5ml/kg/h

● WITH severe malnutrition

- Do not give IV fluids
- Give resomal 5ml/kg every 30 mins for first 2 hours
- Then 5-10mls/kg for the next 4-10 hours
- Give more resomal if child wants more or large stool loss or vomiting
- Check blood glucose and treat if <3mmol/l
- Treat malnutrition as per guidelines

Cardiopulmonary Resuscitation

- Step 1
 - ABCs are priority
 - Check patient responsiveness attempt to rouse the patient with verbal or painful stimuli
 - If no response, assume that the brain is no longer receiving adequate oxygen
- Step 2: Assessment and Management of airway
 - assess for a patent airway, use
 - look/listen/feel approach

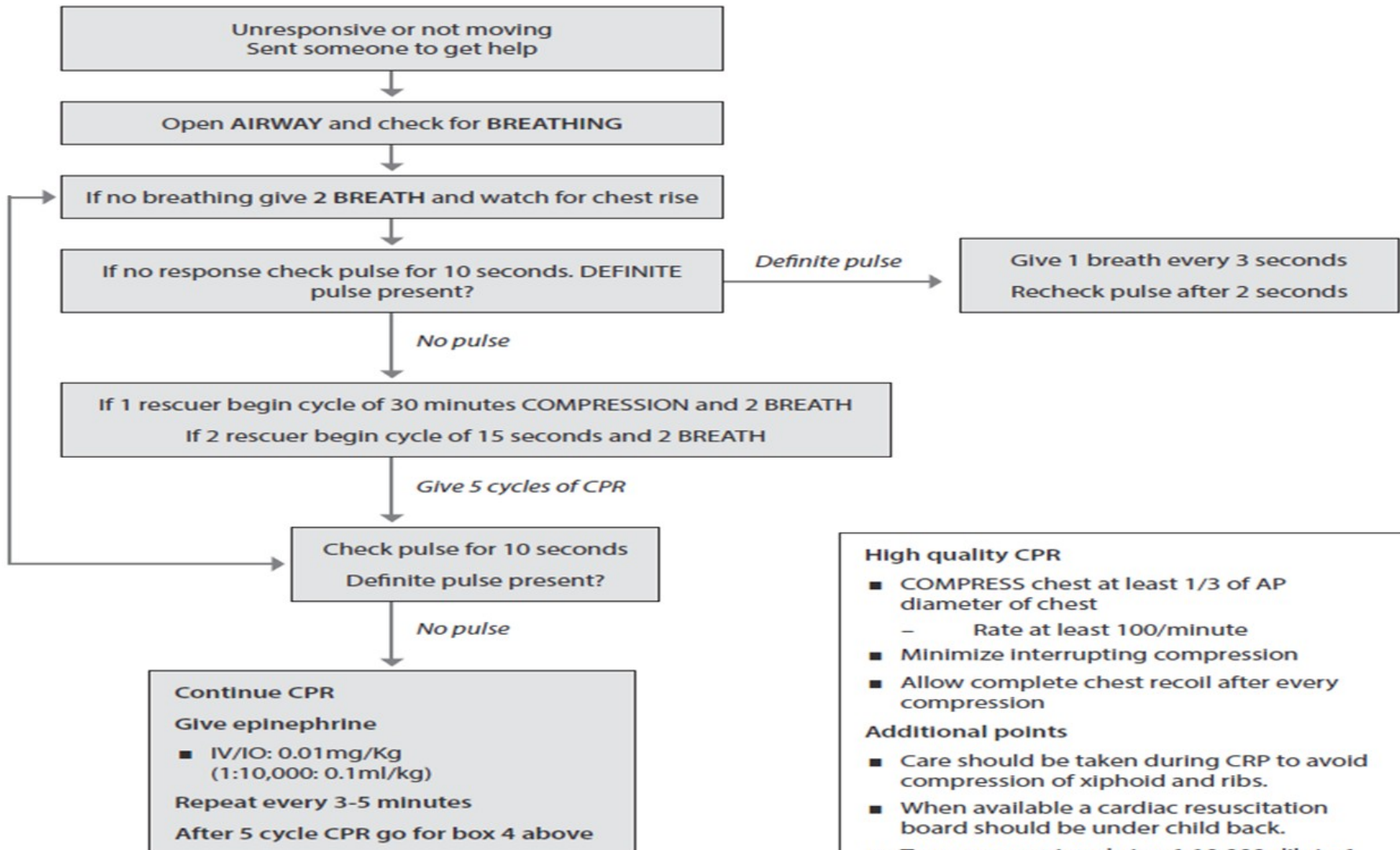
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- Step 3: Giving rescue breaths
 - If the child is not breathing give two breath with a bag mask device or mouth-to-barrier with observing chest rise
 - After two breaths are delivered, assess for a central pulse for 10 sec
- Step 4: Chest compressions
 - palpate the xiphoid process and place hands directly above this area .
 - In older children, place one hand on top of the other In younger children (<8 years) only 1 hand In infants, use 2 fingers in the center of the chest just below the Nipples

Cont'd

- Push the chest, making sure to keep your shoulders directly over the child's chest, push down onto your hands and elbows remain locked
- push fast (rate of 100/min)
- push hard (about 1/3 to 1/2 of AP chest diameter)
- Allow for full chest recoil in between compression and
- Minimize interruptions
- Rotate rescuers ever 2 minutes or sooner when tired

CHART 12. Cardio Pulmonary Resuscitation



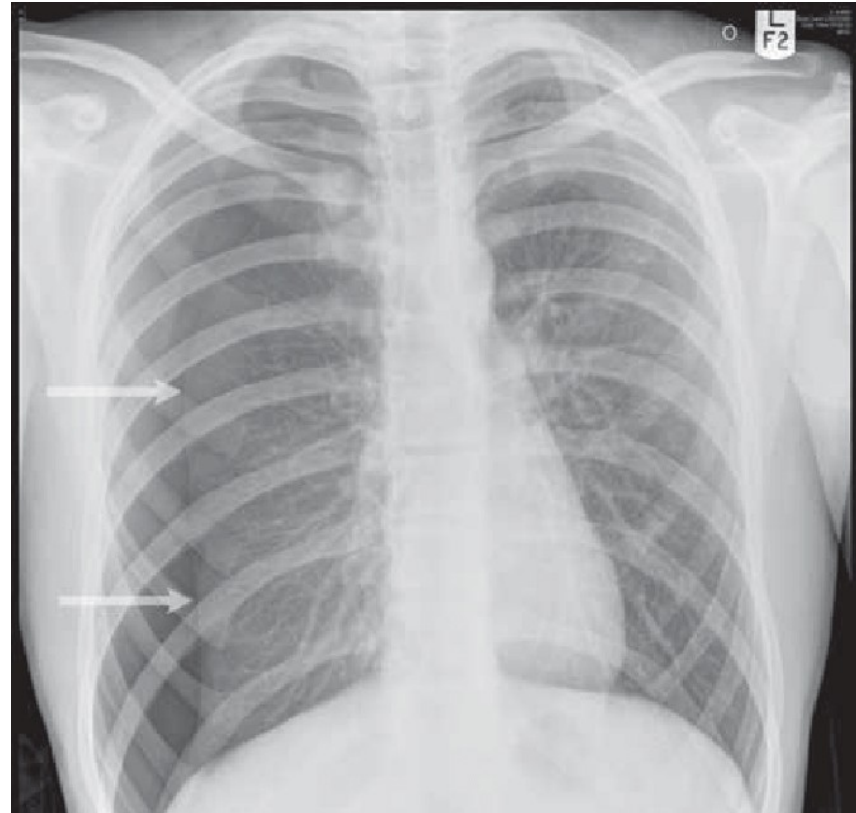
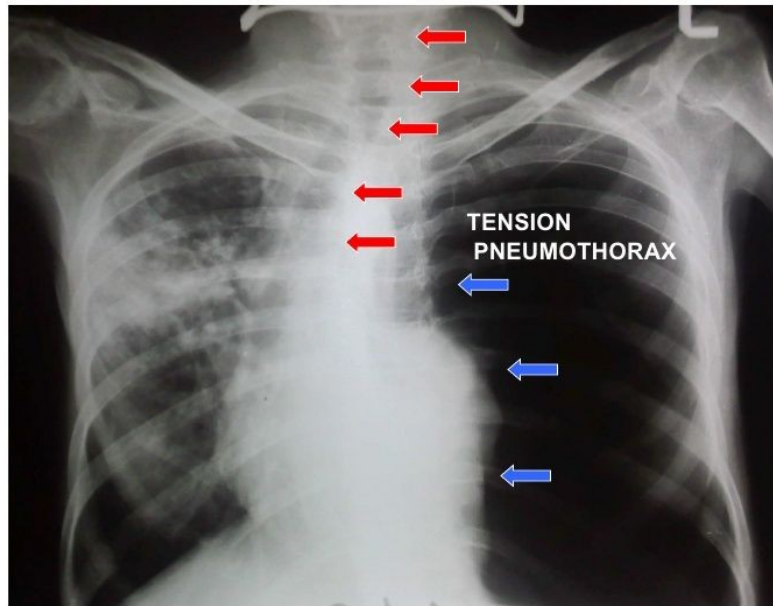
Pediatric Trauma

- **Primary survey and Resuscitation**
 - Airway maintenance with cervical spine protection
 - Breathing and ventilation
 - Circulation with hemorrhage control
 - Disability (evaluation of neurologic status)
 - Other/Exposure and environmental control
- Adjuncts to primary survey
 - Consider laboratory – Hct/Hgb, blood group and Rh, random blood sugar
 - Urinary catheter ;NG tube
 - If ultrasound is available, consider performing FAST u/s
 - have controlled PAIN

Secondary survey

- History (SAMPLE)
 - SAMPLE history
 - S- sign and symptom
 - A - Allergies ? Pencilline
 - M - Medications ? Epileptic, ? Diabetic, asthmatic
 - P - past medical and surgical history
 - L - Time of the child's last meal , helps for OR
 - E - Events preceding the accident Get a good HISTORY
- Complete head to toe evaluation including
 - laboratory studies
 - Radiographic studies

Pneumothorax



- What do you see?
- Differential diagnosis?
- Management?

Cont'd

- What do you see?
 - Collapse of the lungs on the affected side
 - Absence of lung marking
 - Wide intercoastal space
 - If tension pneumothorax
 - Tracheal shift
 - Mediastinal shift
- Differential
 - Chest trauma

Cont'd

- Mx
- Needle decompression in the second intercostal space mid clavicular line
- Chest tube* definitive mx
- Open pneumothorax
 - Three way dressing
 - Chest tube at different site